



LAB 2 • DATA PREPARATION

Transform the data and build the model

Clean by meaning, not appearance. Preserve real negatives and percentages, remove only invalid records, and connect the tables with a predictable star schema.

35–40 minutes Power Query + Model view Deliverable: validated model

Transformation order

Remove invalid rows

Trim + standardize

Apply locale types

Validate counts

Relate tables

Order matters: remove rows that do not belong before spending time converting and correcting their values.

1. Budget vs Actuals

Source text

"(\$120,000)" • 6.00% • N/A

Typed result

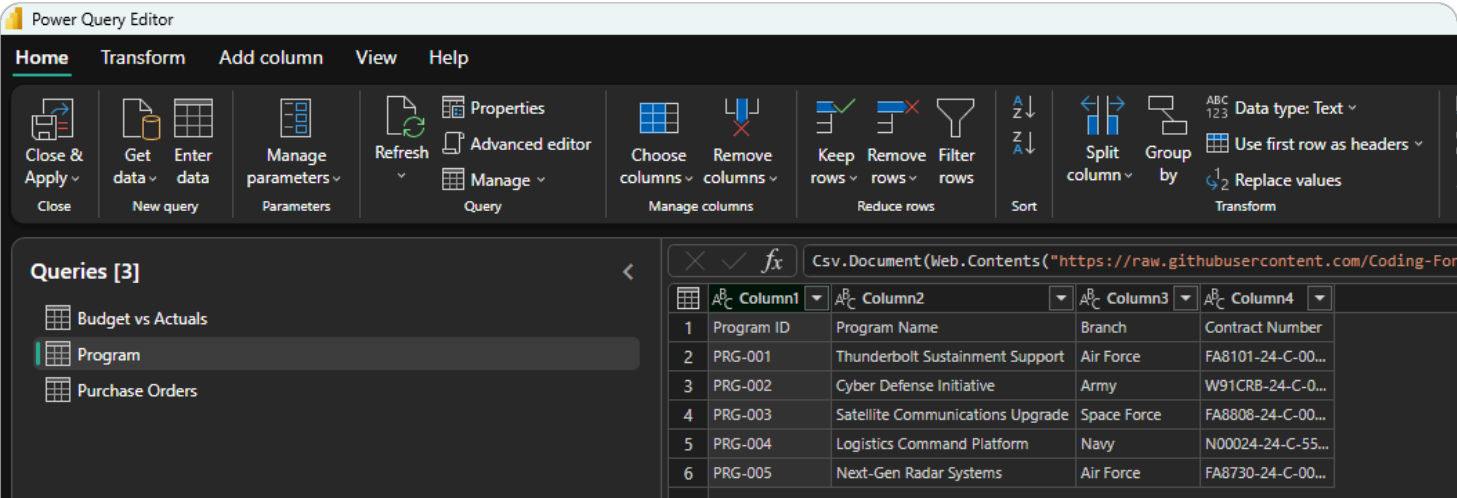
-120000 • 0.06 • null

Promote the source headers

1. If the columns are named Column1, Column2, and so on, select **Home** → **Use first row as headers**.

2. Confirm the first headers are Program ID, Program Name, Fiscal Year, and Quarter.

3. Apply the same step to Program and Purchase Orders if their headers were not promoted during import.



Promote headers before cleaning

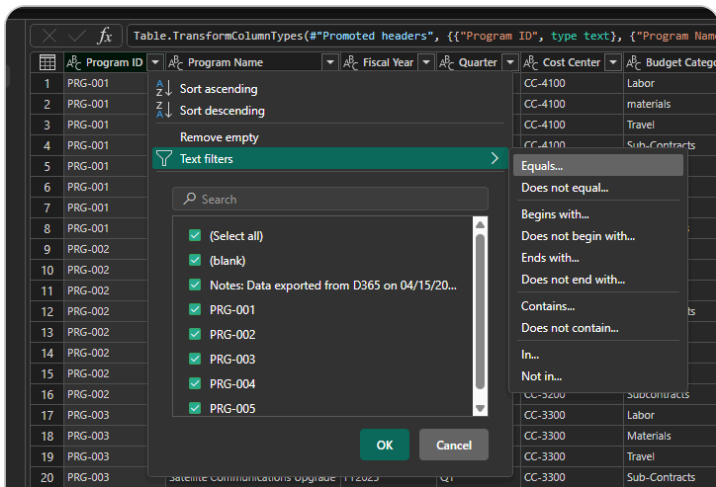
Use the first CSV row as field names before filtering, selecting, or typing columns.

Keep data rows and remove the duplicate

1. Filter **Program ID** with **Text Filters** → **Begins With** → **PRG-**. This safely removes blank and notes rows even if the footer grows.

2. Select all columns and choose **Home** → **Remove Rows** → **Remove Duplicates**.

3. Confirm the query now contains **120 rows**.



Keep valid program records

Use Text Filters → Begins With and enter PRG- rather than deleting a fixed number of footer rows.

Remove the repeated record

Select all columns before choosing Remove duplicates so only an exact duplicate is removed.

Trim and normalize text

1. Select Program ID, Program Name, Cost Center, Budget Category, Funding Source, Contract Number, and Status.
2. Choose **Transform** → **Format** → **Trim**.
3. Capitalize Each Word for Budget Category and Status.
4. In Budget Category, replace Sub-Contracts with Subcontracts.

Quarter	Cost Center	Budget Category	Budgeted Amount	Actual Amount	Variance
	CC-4100	Labor			6.00
	CC-4100	Materials			-6.7
	CC-4100	Travel			20.0
	CC-4100	Sub-Contracts			1.20
	CC-4100	Labor			-56.4
	CC-4100	Materials			2.80
	CC-4100	Travel			-65.7
	CC-4100	Subcontracts			-38.1
	CC-5200	Labor			37.5
	CC-5200	Materials			5.00
	CC-5200	Travel			3.30
	CC-5200	Sub-Contracts			1.30
	CC-5200	Labor			35.0
	CC-5200	Materials			-5.0
	CC-5200	Travel			13.3
	CC-5200	Subcontracts			-1.3
	CC-3300	Labor			1.60
	CC-3300	Materials			-9.1
	CC-3300	Travel			10.0
	CC-3300	Sub-Contracts			-34.7
	CC-3300	Labor	3,100,000.00	\$3,200,000	(\$100,000)
	CC-3300	Materials	2,200,000.00	\$2,100,000	\$100,000

Inspect category values

Open the filter list to confirm which labels still represent the same category.

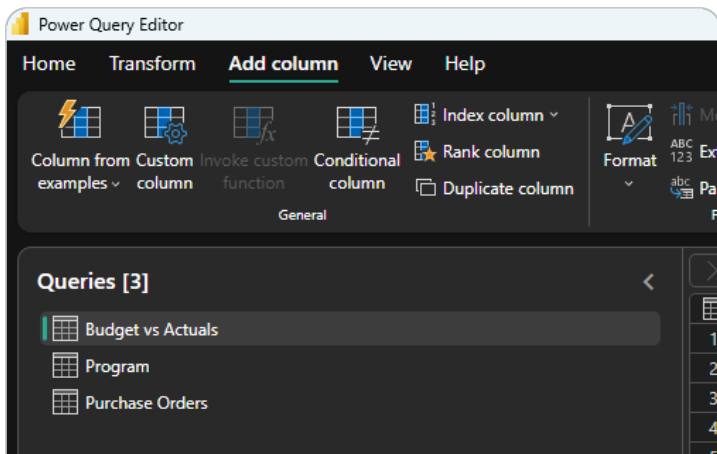
Program Name	Fiscal Year	Quarter	Cost Center	Budget
Thunderbolt Sustainment Support				
Thunderbolt Sustainment Support				
Thunderbolt Sustainment Support				
Thunderbolt Sustainment Support				
Thunderbolt Sustainment Support				
Thunderbolt Sustainment Support				
Thunderbolt Sustainment Support				
Thunderbolt Sustainment Support				
Cyber Defense Initiative				
Cyber Defense Initiative				
Cyber Defense Initiative				
Cyber Defense Initiative				
Cyber Defense Initiative				
Cyber Defense Initiative				
Cyber Defense Initiative				
Cyber Defense Initiative				
Satellite Comm Upgrade				
Satellite Comm Upgrade				
Satellite Communications Upgrade				
Satellite Communications Upgrade				
Satellite Comm Upgrade	FY2025	Q1	CC-3300	Sub-Contract
Satellite Comm Upgrade	FY2025	Q2	CC-3300	Labor

Inspect program-name variants

The source contains shortened and standardized names; the next step creates one governed value for each program.

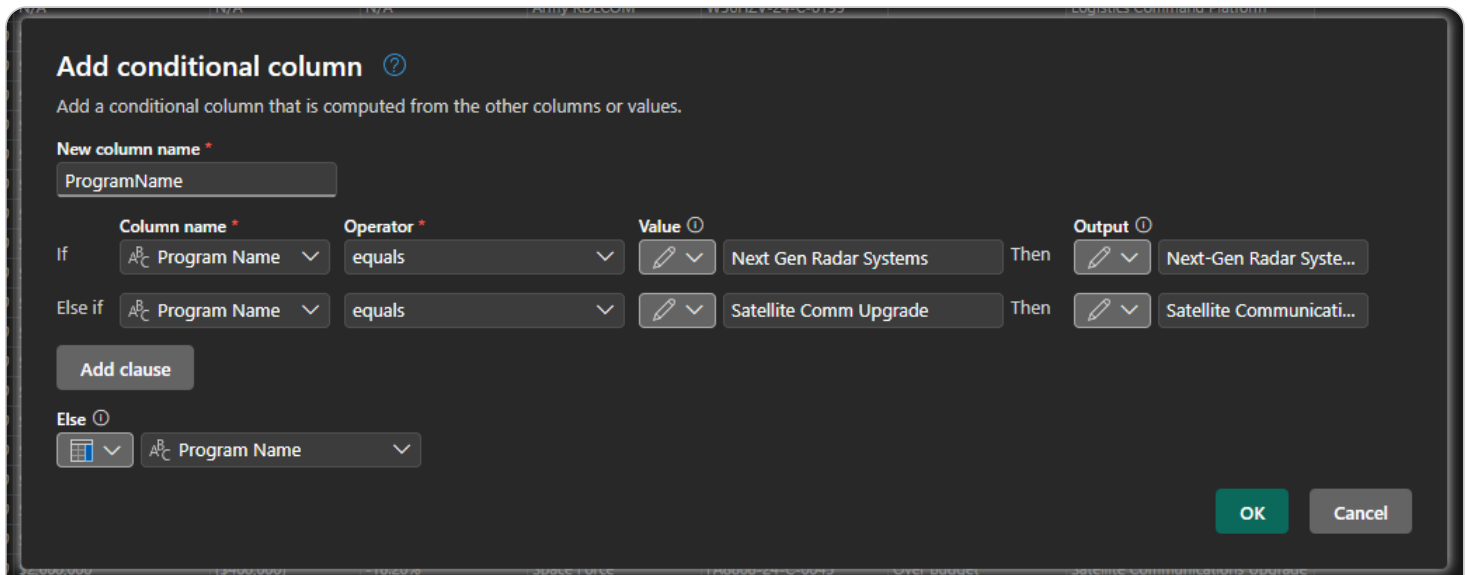
Standardize program names with a conditional column

1. Select **Add column** → **Conditional column**.
2. Name the new column **ProgramName**.
3. Add the rule: if Program Name equals Next Gen Radar Systems, output Next-Gen Radar Systems.
4. Add an Else if rule: if Program Name equals Satellite Comm Upgrade, output Satellite Communications Upgrade.
5. For Else, choose the **Program Name** column so every already-valid name passes through unchanged.
6. Select OK and verify that ProgramName contains only the five governed names.



Open Conditional column

Use Add column → Conditional column to encode both name corrections in one auditable step.



Define the correction rules

Return the standardized name for each known variant and use the original Program Name column as the Else output.

Replace the original name column

1. Right-click the original **Program Name** column and choose **Remove columns**.
2. Rename **ProgramName** to **Program Name**.
3. Set its data type to **Text**.
4. Open the filter list and confirm exactly five program names remain.

able.ReplaceValue("#Inserted conditional column", "Sub-

Program Name	Fiscal Year	Quar
Thunderbolt Sust		Q1
Thunderbolt Sust		Q1
Thunderbolt Sust		Q1
Thunderbolt Sust		Q1
Thunderbolt Sust		Q2
Thunderbolt Sust		Q2
Thunderbolt Sust		Q2
Thunderbolt Sust		Q2
Cyber Defense In		Q1
Cyber Defense In		Q1
Cyber Defense In		Q1
Cyber Defense In		Q1

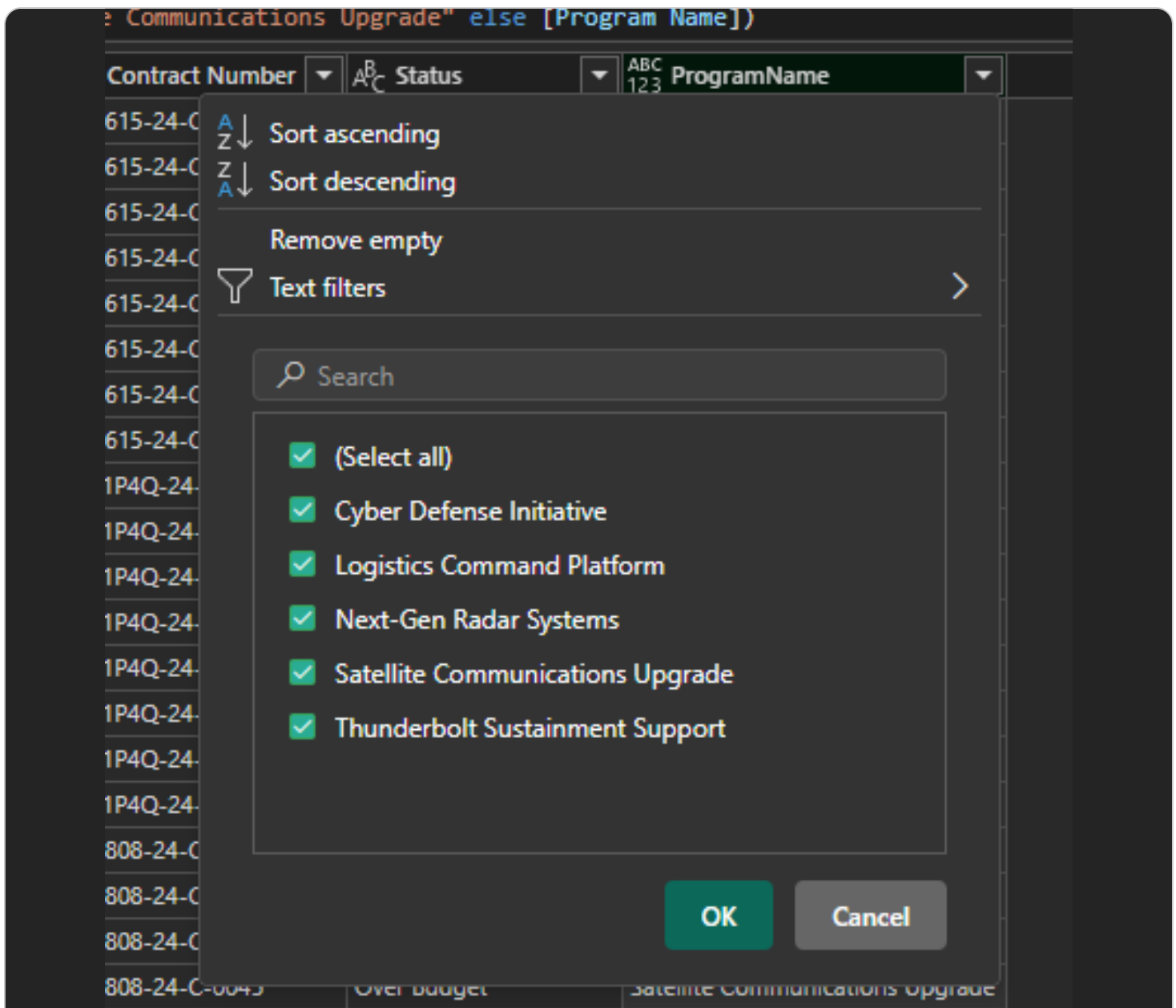
Remove the old column

Delete the inconsistent source column only after the conditional result has been verified.

Status	Program Name
On Track	Thunderbolt Sustainment Support
Over Budget	Thunderbolt Sustainment Support
On Track	Thunderbolt Sustainment Support
On Track	Thunderbolt Sustainment Support
Over Budget	Thunderbolt Sustainment Support
On Track	Thunderbolt Sustainment Support
Over Budget	Thunderbolt Sustainment Support
Over Budget	Thunderbolt Sustainment Support
On Track	Cyber Defense Initiative
On Track	Cyber Defense Initiative
On Track	Cyber Defense Initiative
On Track	Cyber Defense Initiative

Restore the governed field name

Rename ProgramName to Program Name and keep its type as Text.



Verify the result

The final list should contain five names with no shortened Satellite or unhyphenated Next-Gen variants.

Convert money without losing negative values

1. In Actual Amount, Variance, and Variance %, replace N/A with an empty value.
2. Select Budgeted Amount, Actual Amount, and Variance.
3. Choose **Transform** → **Data Type** → **Using Locale**.
4. Use **Currency** and **English (United States)**. Parentheses must remain negative: (\$120,000) becomes -120000.

Convert the percentage correctly

1. Select Variance % and choose **Data Type** → **Using Locale**.
2. Use **Percentage** and **English (United States)**.
3. Verify 6.00% becomes 0.06, displayed as 6% after formatting.

Correction from the old guide: do not remove the percent sign and then convert to Decimal. That produces 6, which Power BI formats as 600%. Also do not strip parentheses without first preserving the negative sign.

Add a reporting period

1. Select **Add Column** → **Custom Column**.
2. Name it **Fiscal Period** and use [Fiscal Year] & " " & [Quarter].
3. This creates sortable labels such as FY2025 Q1 and FY2026 Q2 for the trend chart.

2. Purchase Orders

Deduplicate and trim

1. Select PO ID, then **Home** → **Remove Rows** → **Remove Duplicates**. Expect **20 rows**.
2. Select every text column and apply **Transform** → **Format** → **Trim**.

Clean amount and date types

1. In Amount Paid, replace NULL and - with 0.
2. Select PO Amount, Amount Paid, and Amount Remaining; use **Data Type** → **Using Locale** → **Currency** → **English (United States)**.
3. Select Order Date and Delivery Date; use **Data Type** → **Using Locale** → **Date** → **English (United States)**.
4. Check Column Quality. Do not replace date errors with null until you inspect the original value and confirm it is invalid.

Normalize business labels

1. Capitalize Each Word in PO Status and Priority.
2. Replace Apex Dynamics Corp. with Apex Dynamics Corp and Centurion Inc. with Centurion Inc.
3. Replace Satellite Comm Upgrade with Satellite Communications Upgrade.

3. Program

Protect the dimension key

1. Trim all four columns.
2. Set Program ID, Program Name, Branch, and Contract Number to **Text**.
3. Confirm Program ID is unique: 5 rows and 5 distinct values.

Modeling rule: use **Program[Program Name]** in slicers and axes. The fact-table names were intentionally inconsistent; the dimension is the governed label.

Validate before loading

120
Budget rows

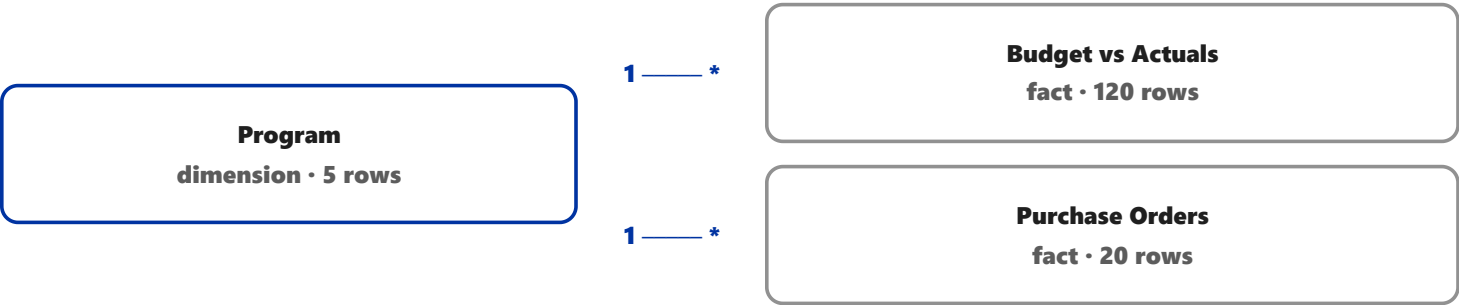
20
PO rows

5
Programs

0
Errors

- 1. Set Column Profiling to the entire dataset.
- 2. Confirm amount columns show numeric distributions, not text values.
- 3. Confirm Variance % is Percentage and dates are Date.
- 4. Select **Home** → **Close & Apply**.

Build the star schema



- 1. Open **Model view**.
- 2. Relate **Program[Program ID]** to **Budget vs Actuals[Program ID]**.
- 3. Relate **Program[Program ID]** to **Purchase Orders[Program ID]**.
- 4. For both relationships use **One-to-many (1:*)**, **Single** cross-filter direction, and make the relationship active.
- 5. Hide Program Name in both fact tables to steer report authors toward the Program dimension.

Single-direction filters flow from Program to both facts. This lets one Program slicer filter budget and procurement without ambiguous loops.